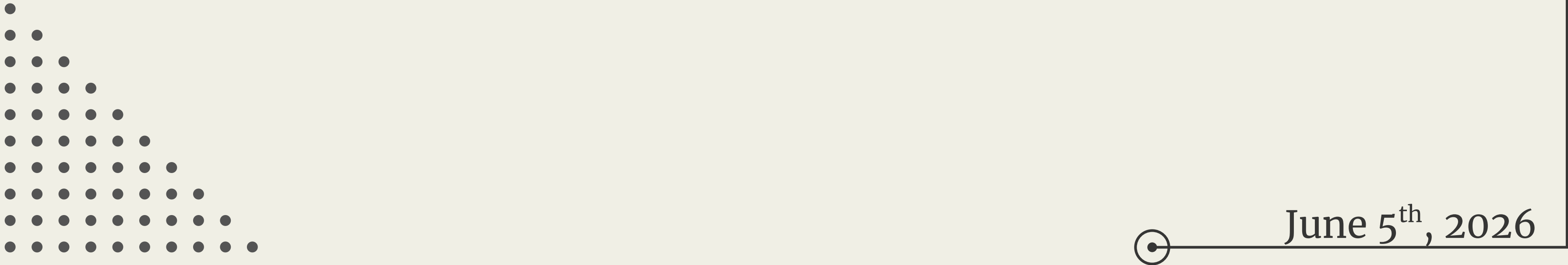
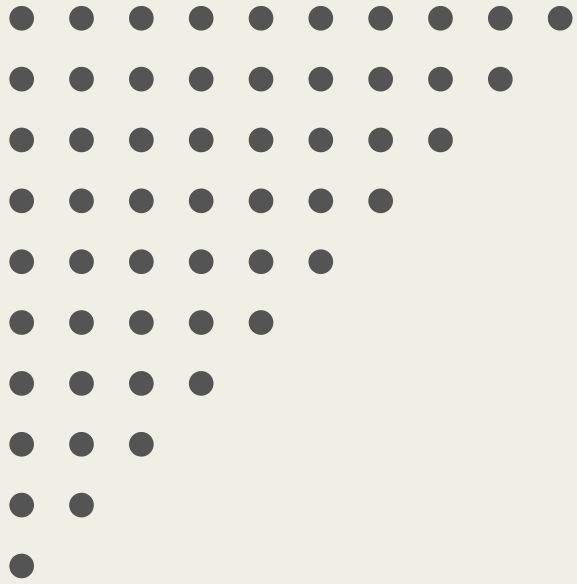


TODO APPLICATION

by Daria Ruchala

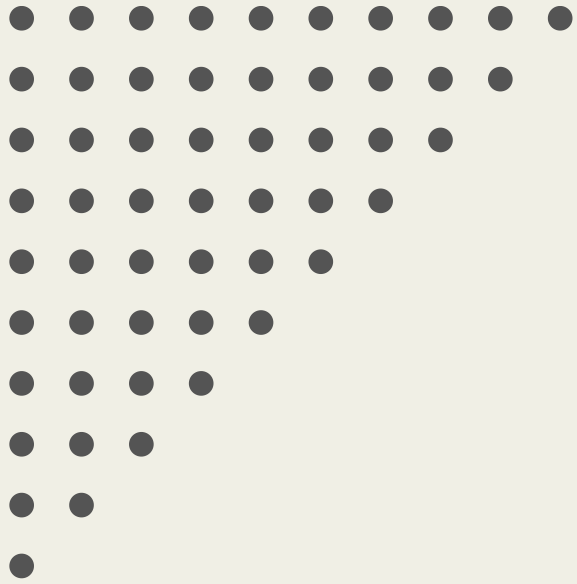


June 5th, 2026



OVERVIEW

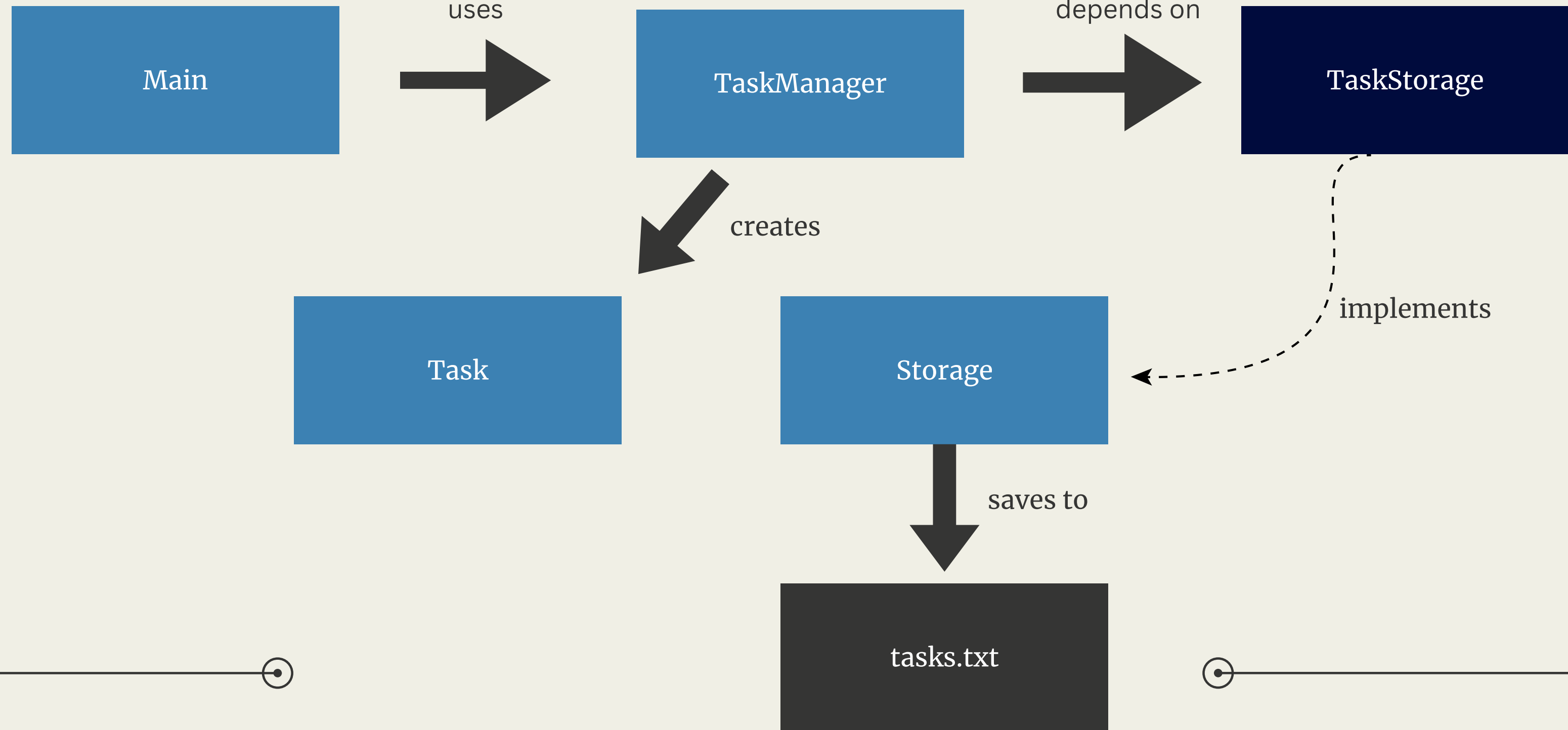
- Java application
- CLI based
- Users add, remove, status update , and filter tasks
- persistence between tasks session to session due to task.txt

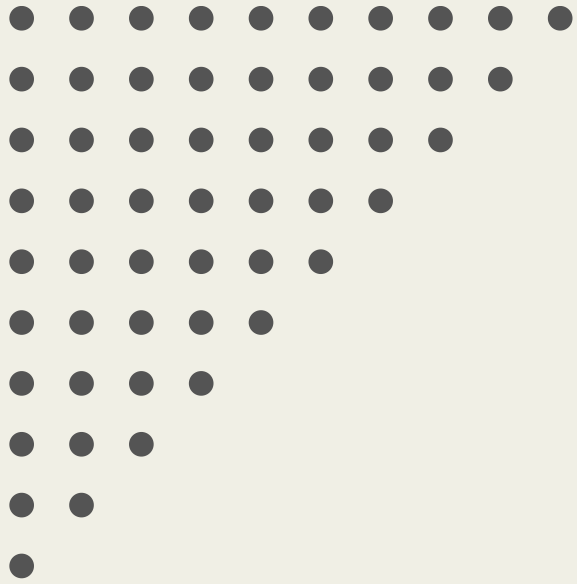


APPROACH AND DESIGN

- 4 classes and 1 interface
- Task: encapsulates all task data (id, title, category, status)
- TaskStorage: interface that defines how tasks are saved and loaded
- Storage: implements TaskStorage, saves tasks to tasks.txt
- TaskManager: handles all the business logic
- Main: entry point, bridges the user and application logic

ARCHITECTURE





OOP CONCEPTS

- Encapsulation: Task fields are private, accessed only through getters and setters
- Abstraction: TaskStorage interface hides the details of how tasks are stored
- Single Responsibility Principle: each class has one job and one job only



HOW TO RUN

Requirements: Java 25 or higher

01

02

03

04

Open terminal and navigate
to the project folder

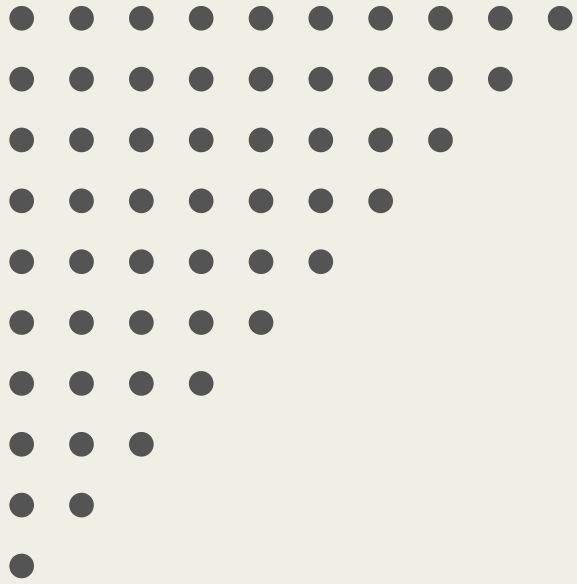
Compile:
`javac src/*.java -d out`

Run:
`java -cp out Main`

Follow menu prompts



05



HOW TO TEST

01

Compile:

`javac src/*.java -d out`

02

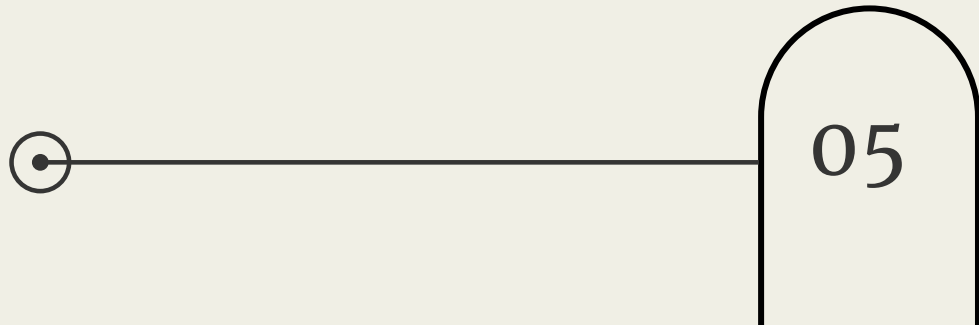
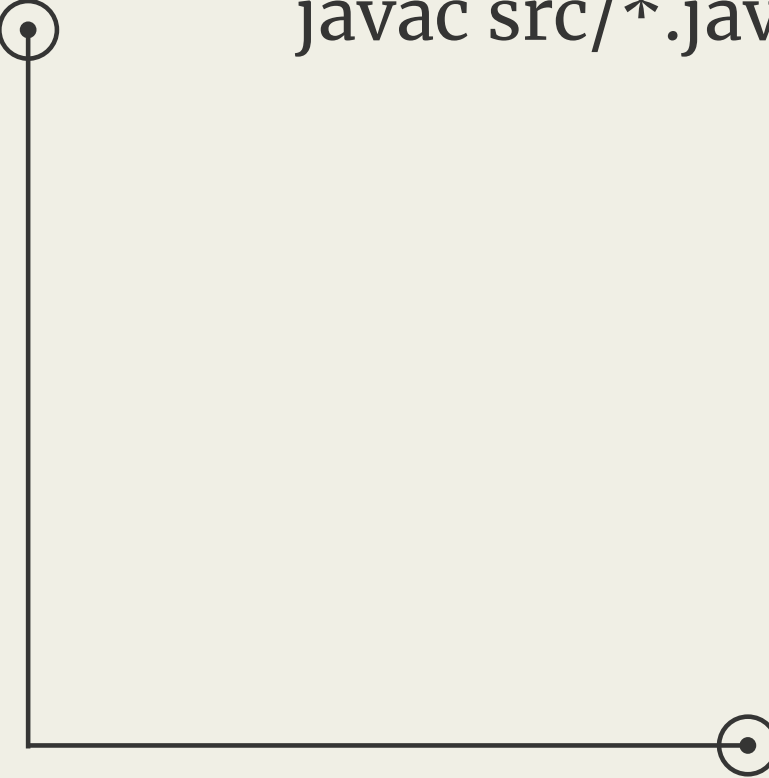
Run the tests:

`java -cp out
TaskManagerTest`

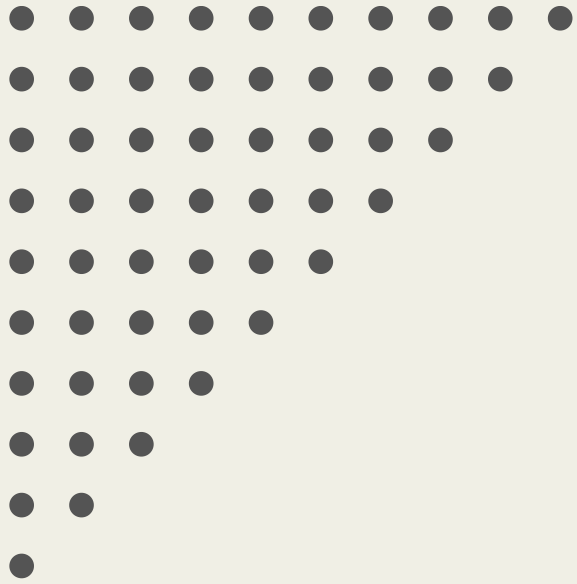
03

Expected output:

`testAddTask PASSED
testRemoveTask PASSED
testMarkDone PASSED
testListByCategory PASSED
testSaveAndLoad PASSED`



05



SEED DATA

Tasks are saved in tasks.txt in the following format:
(id,title,category,status)

Sample data included:

1,clean,home,false

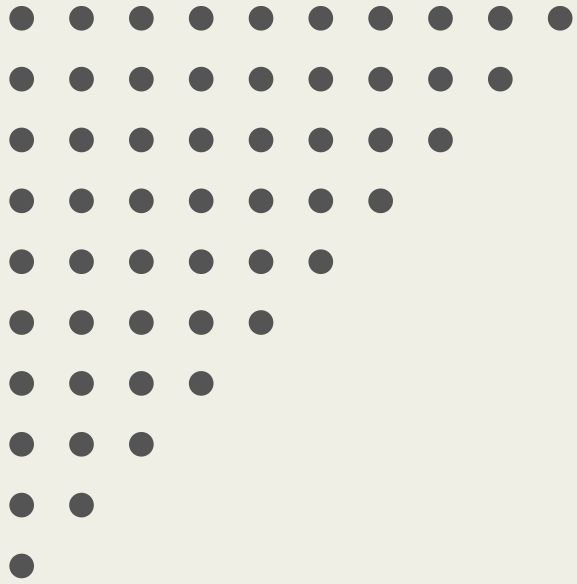
2,clean,home,true

3,gym,personal,false

4,buy groceries,shopping,false

5,cook,personal,false

false = pending, true = done

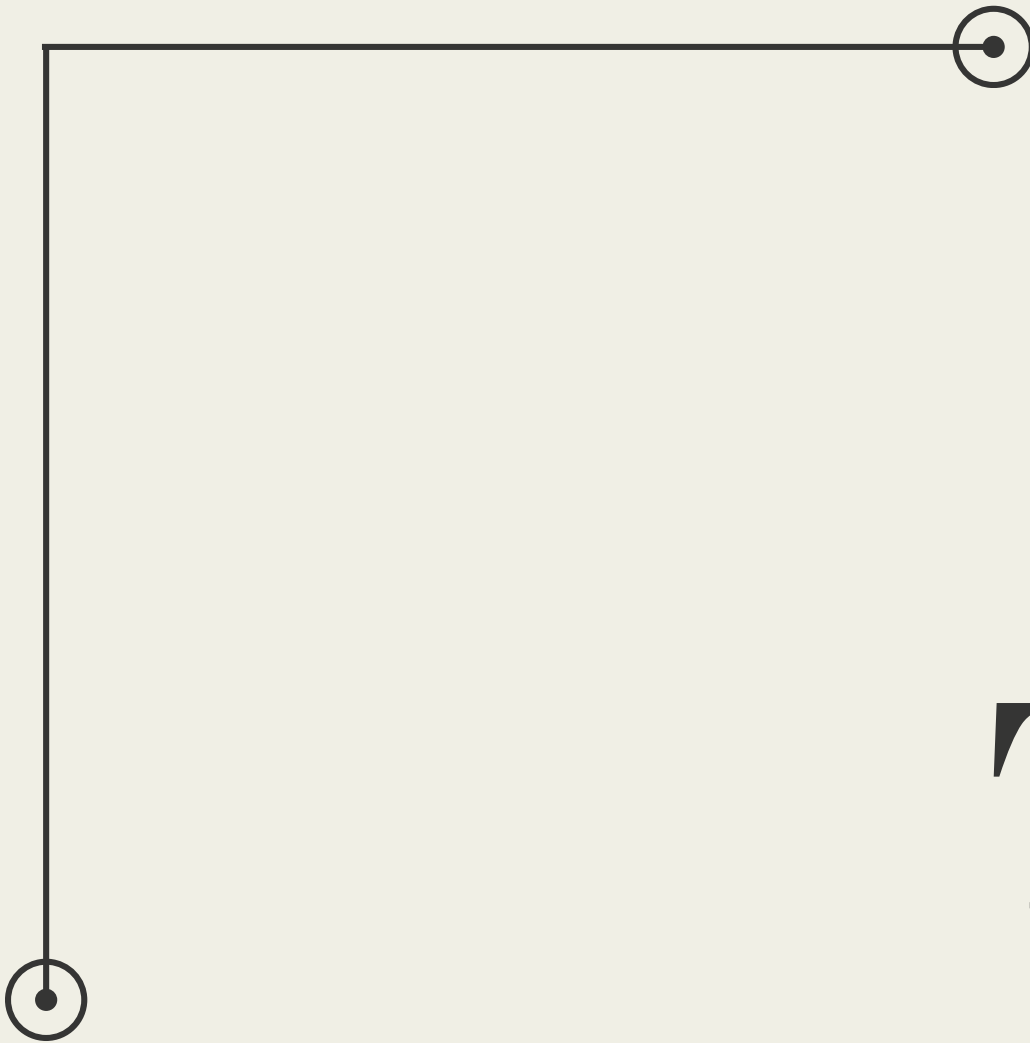


TEST DATA

- testAddTask: adds "Test Task" to "Work" category, verifies task count equals 1
- testRemoveTask: adds a task then removes it, verifies task count equals 0
- testMarkDone: adds a task and marks it done, verifies task count equals 1
- testListByCategory: adds 3 tasks across 2 categories, verifies total count equals 3
- testSaveAndLoad: saves a task, reloads from file, verifies task persisted correctly

CODE COVERAGE

Element ^	Class, %	Method, %	Line, %	Branch, %
✓ all	80% (4/5)	86% (20/23)	52% (71/135)	22% (11/50)
🔗 Main	0% (0/1)	0% (0/1)	0% (0/37)	0% (0/14)
🔒 Storage	100% (1/1)	100% (3/3)	56% (9/16)	50% (2/4)
🔒 Task	100% (1/1)	100% (6/6)	100% (10/10)	100% (0/0)
🔒 TaskManager	100% (1/1)	75% (6/8)	57% (22/38)	20% (5/24)
🔗 TaskManagerTest	100% (1/1)	100% (5/5)	88% (30/34)	50% (4/8)
🔒 TaskStorage	100% (0/0)	100% (0/0)	100% (0/0)	100% (0/0)



THANK YOU

by Daria Ruchala

